

2023

Q1.

(a) To be in saturation: $V_{DS} > V_{GS} - V_{th}$, $V_{DS} > V_{GS} - V_{th}$

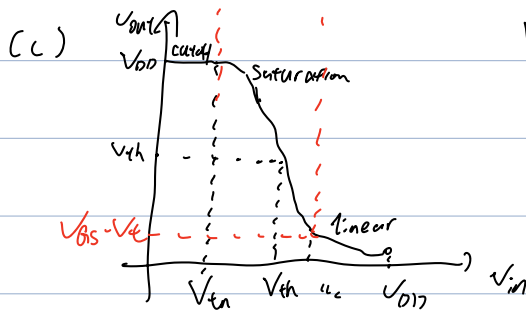
When it just reach this condition: $V_{DS} = V_{GS} - V_{th}$

$$\Rightarrow I_{DS} = \frac{\beta}{2} [2(V_{GS} - V_{th})V_{DS} - V_{DS}^2] = \frac{\beta}{2} \cdot V_{DS}^2$$

(b) When $V_{GS} = 1.2V$,

$$V_{DS, min} = V_{GS} - V_{th} = 1.2 - 0.5 = 0.7 \checkmark$$

$$I_{DS} = \frac{\beta}{2} V_{DS, min}^2 = \frac{20\mu}{2} \cdot 0.7^2 = 4.9\mu A$$



$$V_{out} = V_{in} - V_{th} = V_{in} - I_D R_D$$

$$I_D = \frac{\beta}{2} (V_{in} - V_{th})^2$$

$$V_{in} - V_{th} = V_{DD} - \frac{\beta}{2} (V_{in} - V_{th})^2 R_D$$

$$\Rightarrow \frac{\beta R_D}{2} (V_{in} - V_{th})^2 + (V_{in} - V_{th}) - V_{DD} = 0$$

$$\Rightarrow V_{in} - V_{th} = \frac{-1 + \sqrt{1 + 2\beta R_D V_{DD}}}{\beta R_D}$$

$$\Rightarrow V_{in} = V_{th} + \frac{-1 + \sqrt{1 + 2\beta R_D V_{DD}}}{\beta R_D}$$

The Saturation region.

This region can provide highest gain.

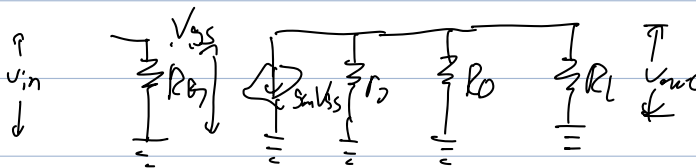
high output resistance

Q2.

(a). Common source amplifier.

By cascading another CS amplifier using this transistor.

(b). The small signal equivalent circuit:



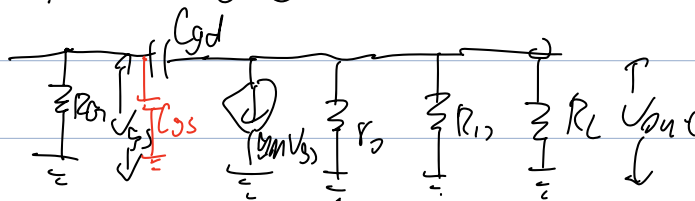
Assume: ideal transistor working in saturation

AC < DC and DC chort to ground for AC

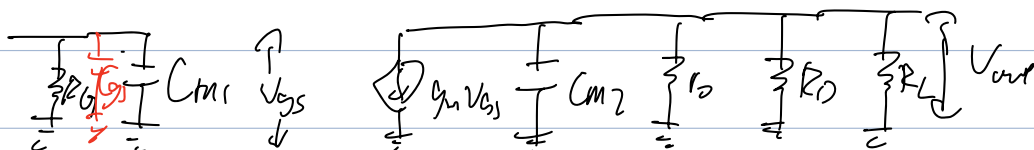
mid-band frequency: $\left\{ \begin{array}{l} \text{External capacitor short} \\ \text{internal capacitor open} \end{array} \right.$

(c). When frequency higher than mid band:

before applying Miller's Theorem,



After applying:



$$C_{M1} = \frac{C_{gd}}{1+A} \quad C_{M2} = \frac{C_{gd}}{1+1/A}$$

$$A = -g_m(r_o \parallel R_D \parallel R_L)$$

Miller's Theorem can help us to analysis high-frequency behavior when we need to consider the internal capacitor due to its non-ideality.

Changes: Add C_{gd} . Separate it into C_{M1} and C_{M2} after Miller's theorem.

Q3

Cal. Figure 3(a): Current mirror

Figure 3(b): Wilson Current mirror.

Figure 3(c): Cascode current mirror

The 3(c) has the best accurate current balance.

Another advantage: ^{non} One pure impedance

disadvantage: uses the most transistor.

requiring largest space, and more money.

need higher drain voltage

cb). For m_1 : $I_{in} = \frac{\mu_1}{2} \frac{W_1}{L_1} (V_{GS1} - V_{T1})^2$

For m_2 : $I_{out} = \frac{\mu_2}{2} \frac{W_2}{L_2} (V_{GS2} - V_{T2})^2$

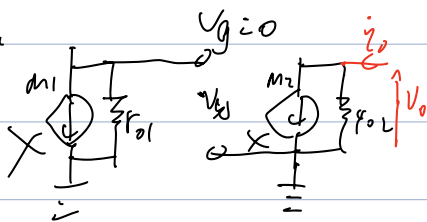
$V_{GS1} = V_{GS2}$

Assume they have the same fabrication process, i.e.,

$\mu_1 = \mu_2$, $V_{T1} = V_{T2}$, no channel length modulation

Therefore, $\frac{I_{out}}{I_{in}} = \frac{W_2/L_2}{W_1/L_1} = \frac{W_2 L_1}{W_1 L_2}$

(c).

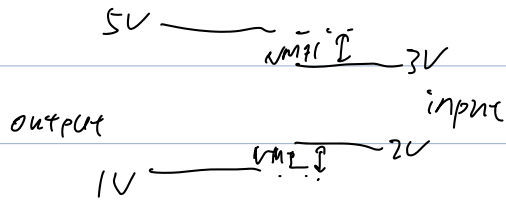


By using inspection mode,

canceling all input source, i.e., V_{g20}

$R_o = r_{o2}$

Q4. (a)



Therefore $NMFI = V_{DD} - V_{th} = 2V$

$NML = V_{th} - V_{OL} = 1V$

(b). When output is high, static power dissipation occurs,

$$I_D = \frac{V_{DD} - V_{OL}}{R_D} = \frac{5 - 1}{2k} = 2mA$$

$$P_{static} = V_{DD} I_D = 5 \times 2m = 10mW$$

To reduce the static power dissipation,

we can simply increase R_D .

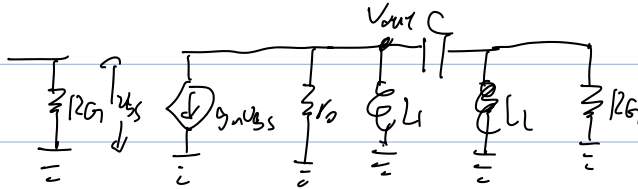
However, this will also increase the rising time.

(c). No static power dissipation, and faster switching speed.

Q5.

(a). Hartely oscillator

Small signal equivalent circuit.



$$R_G = R_1 // R_2$$

Assume: ideal transistor working in saturation.

AC $\ll$ DC and V_{GS} short to ground for DC

mid-band frequency: $\left\{ \begin{array}{l} \text{external capacitor short} \\ \text{internal capacitor open} \end{array} \right.$

(b). For oscillation to happen: $AB \geq 1 + j0$

$|AB| \geq 1$ and phase $n(2\pi)$, $n=0,1,2,\dots$

Barkhausen criterion.

(c). From Barkhausen criterion:

$$R_G(\omega L_1 - \frac{1}{\omega C} + \omega L_2) = 0$$

$$\Rightarrow \omega = \sqrt{\frac{1}{C(L_1 + L_2)}}$$

$$f_0 = \frac{1}{2\pi\omega} = \frac{1}{2\pi} \cdot \frac{1}{\sqrt{C(L_1 + L_2)}}$$

(d). To let $|AB| \geq 1$ where $\omega = \frac{1}{\sqrt{C(L_1 + L_2)}}$

$$AB = \frac{g_m R_G \omega^2 L_1 L_2}{-\omega^2 L_1 L_2 + \frac{L_2}{C}} = \frac{g_m R_G \cdot \frac{L_1 L_2}{C(L_1 + L_2)}}{-\frac{1}{C} \cdot \frac{L_1 L_2}{L_1 + L_2} + \frac{L_2}{C}} \gg 1$$

$$\Rightarrow g_m > \frac{L_2}{R_G L_1}$$

Therefore, the minimum $g_{m-\min} = \frac{L_2}{R_G L_1}$

, which only depends on R_G, L_1, L_2

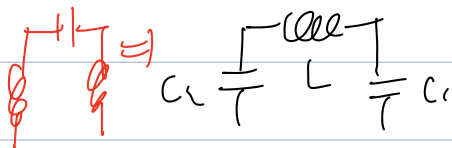
$$R_G = R_1 // R_2 = 5 \text{ k}\Omega$$

$$g_{m-\min} = \frac{L_2}{R_G L_1} = \frac{10 \text{ nH}}{5 \text{ k}\Omega \cdot 2 \text{ mH}} \approx 1 \text{ m A/V}$$

$$f_0 = \frac{1}{2\pi} \cdot \frac{1}{\sqrt{C(L_1 + L_2)}} = \frac{1}{2\pi} \cdot \frac{1}{\sqrt{10 \text{ nF} \cdot (2 \text{ mH} + 10 \text{ nH})}} \approx 1.02 \text{ MHz}$$

(e) Colpitts oscillator.

The load is changed by 2 capacitors and one inductor like following:



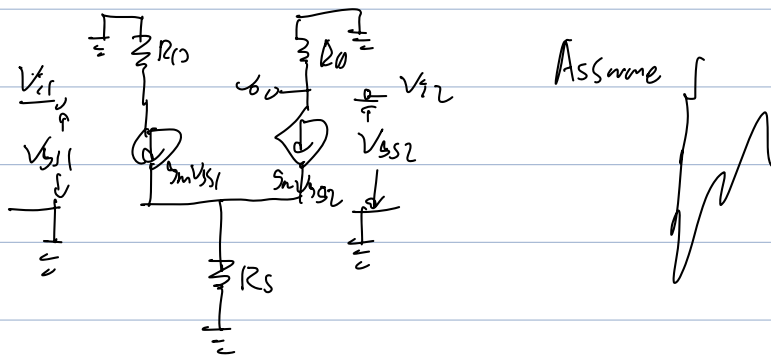
Qb.

cn1. Long tail ^{pair} single ended differential amplifier

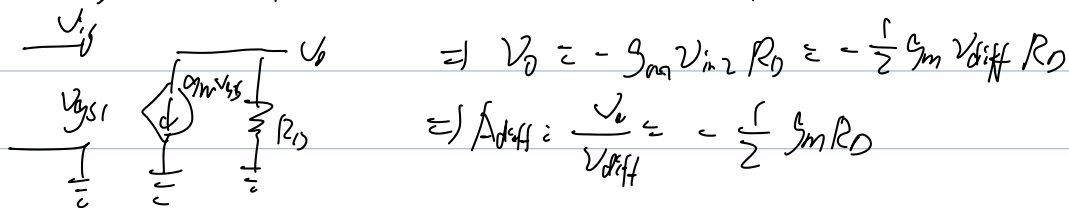
role: work in the 1st stage to

amplify the differential signal,
~~decreasing~~ ^{high CMRR} common mode signal,
~~reject~~
 providing large input impedance.

cn1.



To get A_{diff} , the above can be transformed as below.



$$(c) A_{diff} = -\frac{1}{2} g_m (r_{o6} || r_{o2})$$

M_5, M_6 can be considered as effective resistor r_{o5}, r_{o6} , respectively. Can no longer ignore r_{o1} , since r_{o6} is big enough.

$A_{cm} \approx -\frac{r_{ob}}{2r_o}$ which also changed by replacing R_s to its effective resistance r_{os} .

For differential signal, they destructively interfere and form virtual ground so do not pass through $M3$.

For common mode signal, they add and pass through $M3$

reduce CM gain, increase Diff gain. *Unhelpful*

(d). For A_{CL} they enables higher output gain

For D.C., they allows us to get higher output impedance without reducing current and bias voltage.

① set the DC current level

② high impedance and high current at the same time